

Section One: Calculator-free

35% (52 Marks)

This section has eight (8) questions. Answer all questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1

(6 marks)

Michael is planning on buying a boat priced at \$20 000 from a shop called Magical Marine. The shop offers a special loan package where the buyer pays a 10% deposit, then pays the remaining balance and interest in monthly installments for 4 years. Magical Marine charge 15% p.a. simple interest on all finance packages.

(a) Calculate the deposit Michael will pay on the boat.

(1 mark)

$$\begin{aligned}\text{Deposit} &= 10\% \text{ of } \$20\,000 \\ &= \$2\,000\end{aligned}$$

- Calculates deposit. ✓

(b) Calculate the remaining balance Michael will owe the boat shop.

(1 mark)

$$\begin{aligned}\text{Remaining Balance} &= \$20\,000 - \$2\,000 \\ &= \$18\,000\end{aligned}$$

- Calculates Remaining Balance ✓

(c) Calculate the interest Michael must pay on the remaining balance.

(2 marks)

$$\begin{aligned}\text{Interest for 1 year} &= 15\% \text{ of } \$18\,000 \\ &= \$2\,700\end{aligned}$$

- Calculates I for 1 year ✓

$$\therefore \text{Interest for 4 years } \$2\,700 \times 4 = \$10\,800$$

- Calculates I for 4 years ✓

(d) Calculate the total amount Michael will repay Magical Marine.

(1 mark)

$$\begin{aligned}\text{Total Repayment} &= \text{Interest} + \text{Remaining Balance} \\ &= 10\,800 + 18\,000 = \$28\,800\end{aligned}$$

- Calculates total amount to be repaid. ✓

(e) Write an expression that would calculate Michael's monthly repayment.

(1 mark)

$$\frac{28\,800}{48}$$

★ → (48 months in 4 yrs) repayment.

- Writes expression to calculate monthly repayment. ✓

See next page

Question 2

(6 marks)

Evaluate

(a) $4 \times 20 - 10 \div 5 + 15.$

$= 80 - 2 + 15$

$= 93$

★ Calculate $80 - 2 + 15$

from left to right.

★ $80 - 2 + 15 \neq 80 - 17$

B
1M } same level of priority, calculate left to right (2 marks)
D }A
5

(as above)

Indicates use of rule ✓

of order

Calculates correct value ✓

(b) $2a^2 - 6a$ when $a = \frac{1}{2}.$

(2 marks)

$= 2\left(\frac{1}{2}\right)^2 - 6\left(\frac{1}{2}\right)$

$= 2\left(\frac{1}{4}\right) - 3$

$= \frac{1}{2} - 3$

$= -2\frac{1}{2}$

★ put brackets around $\frac{1}{2}$ when raising to the power of 2 because

$\left(\frac{1}{2}\right)^2 = \frac{1}{4}$ whilst $\frac{1^2}{2} = \frac{1}{2}$

- Substitutes correctly ✓

- Calculates correct value ✓

(c) $p^2 + pq + \sqrt{q}$ when $p = -2$ and $q = 16.$

(2 marks)

$= (-2)^2 + (-2)(16) + \sqrt{16}$

$= 4 - 32 + 4$

$= -24$

★ $(-2)^2 = -2 \times -2$
 $= 4$

Different to $-2^2 = -2 \times 2$
 $= -4$

- Evaluates at least two of the three terms correctly. ✓

- Calculates correct value ✓

See next page

Question 3

(9 marks)

Four matrices are defined as $A = \begin{bmatrix} 4 \\ -1 \\ 6 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 & -1 \\ 7 & -2 & 3 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 3 \\ -2 & 0 \end{bmatrix}$ and $D = [5 \quad -4]$.

(a) State

(i) which matrix has more rows than columns.

(1 mark)

Matrix A

- Identifies correct matrix ✓

(ii) the size of the row matrix.

(1 mark)

Matrix D is a 1×2

- Identifies correct dimensions ✓

★ Row matrix has 1 row

(b) Calculate $5C + 3I$, where I is the 2×2 identity matrix.

(3 marks)

$$5 \begin{bmatrix} 1 & 3 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} 5 & 15 \\ -10 & 0 \end{bmatrix}$$

- Calculates $5C$ ✓

- Calculates $3I$ ✓

- Calculates correct sum ✓

$$\therefore 5C + 3I = \begin{bmatrix} 5 & 15 \\ -10 & 0 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 8 & 15 \\ -10 & 3 \end{bmatrix}$$

(c) Calculate C^2 .

(2 marks)

$$\begin{bmatrix} 1 & 3 \\ -2 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 3 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} -5 & 3 \\ -2 & -6 \end{bmatrix}$$

- Calculates one correct row or column ✓

- Calculates correct product ✓

★ perform matrix multiplication to square a matrix. Don't just square each element.

(d) It is required to multiply all four matrices together so that the final product is a square matrix with the largest possible number of columns. State the order in which the matrices should be multiplied and state the number of columns in the final product.

(2 marks)

Order: $A \times D \times C \times B$

- States order that produces a square matrix ✓

Product will have 3 columns.

- States correct number of columns ✓

See next page

Question 4

(6 marks)

During the week, casual employees at a recycling plant are paid a wage of \$25 per hour plus \$1.50 for every kg of electronic waste they collect. Over the weekend, both rates are increased by 20%.

A casual worked for 4 hours on Thursday, collecting 30 kg of electronic waste, and then a further 6 hours on Sunday when they collected 40 kg of electronic waste.

(a) Determine the hourly pay rate for weekends.

(1 mark)

$$25 \times 0.2$$

$$= 5$$

$$\therefore \text{New pay rate} = \$30 \text{ per hour}$$

- Calculates correct rate ✓

(b) Calculate how much the casual earned on Thursday.

(2 marks)

$$\text{Wage} = 4 \times 25$$

$$= \$100$$

$$\text{Waste Wage} = 30 \times 1.5$$

$$= \$45$$

$$\text{Total Earnings for Thursday} = \$145$$

- Calculates electronic waste collection earnings ✓

- Calculates standard wages and adds to calculate total earnings ✓

(c) Calculate the total earned by the casual over the two days.

(3 marks)

Sunday Earnings.

$$\text{Wage} = 6 \times 30$$

$$= 180$$

$$\text{Waste Wage} = 40 \times 1.5$$

$$= 60$$

$$60 \times 0.2$$

$$= 12$$

$$\therefore \text{Waste Wage}$$

$$60 + 12 = 72$$

$$\text{Total Sunday Earnings}$$

$$= 180 + 72$$

$$= \$252$$

Total for 2 days

$$252 + 145$$

$$= \$397$$

- Calculates Sunday total ✓

- Increases total by 20% ✓

- Calculates correct for 2 days ✓

★ Set working out clearly with subheadings

See next page

Question 5

(6 marks)

Hockey teams M and N had each played 18 games in a league. Team M had won 5, lost 7 and drawn the remainder of its games. In total, the two teams had won 11 and drawn 16 games.

- (a) Show this information in a fully labelled 2×3 matrix A , with columns ordered win, draw, loss and rows ordered team M and team N. (2 marks)

$$A = \begin{matrix} & \begin{matrix} W & D & L \end{matrix} \\ \begin{matrix} M \\ N \end{matrix} & \begin{bmatrix} 5 & 6 & 7 \\ 6 & 10 & 2 \end{bmatrix} \end{matrix}$$

- Finds one correct row ✓
- Shows correct matrix ✓

★ Read the Q: Columns ordered win, draw, loss
Rows ordered M, N
Matrix labelled A

- (b) Each team gains 5 points for a win, 2 points for a draw and no points for a loss. Show this information in matrix B that can be multiplied by matrix A . (1 mark)

$$B = \begin{matrix} & \begin{matrix} W & D & L \end{matrix} \\ \begin{matrix} W \\ D \\ L \end{matrix} & \begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix} \end{matrix}$$

★ label matrix

- Shows correct matrix ✓

- (c) Calculate $A \times B$ and state which team is ahead in the league and by how much. (3 marks)

$$\begin{bmatrix} 5 & 6 & 7 \\ 6 & 10 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix} = \begin{matrix} M \\ N \end{matrix} \begin{bmatrix} 37 \\ 50 \end{bmatrix}$$

- Obtains a 2×1 product ✓
- calculates correct product ✓
- Appropriate interpretation ✓

Team N is ahead in the league by 13 points.

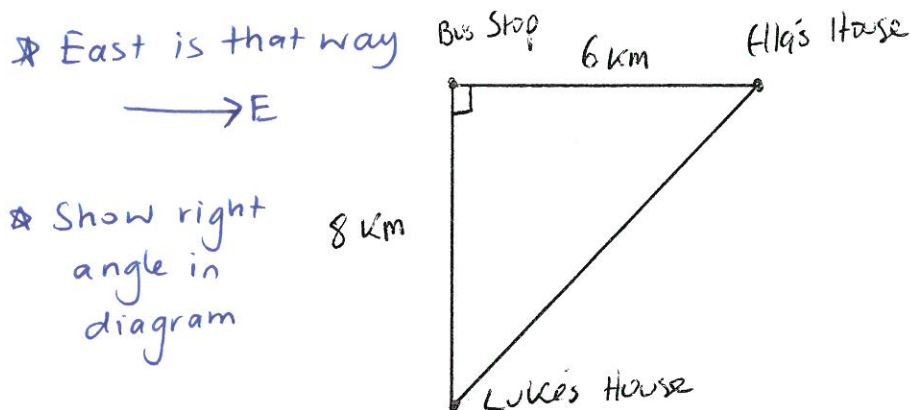
Question 6

(7 marks)

Jimmy walked 6km east from the bus stop to Ella's house. Luke walked 8km south from the same bus stop to his house.

- (a) Draw a labelled diagram to show the locations of the two houses and the bus stop.

(2 marks)



- Correct orientation of diagram and houses.
distances labelled. ✓

- Right angle included ✓

- (b) Determine the shortest distance from Luke's house to Ella's house.

(2 marks)

$$\begin{aligned} x &= \sqrt{8^2 + 6^2} \\ &= \sqrt{64 + 36} \\ &= 10 \end{aligned}$$

∴ Shortest distance between houses is 10 km.

- Uses Pythagorean theorem. ✓

- Calculates shortest distance between houses ✓

Imagine Ella walked to Luke's house via the bus stop.

- (c) How much further would she have travelled compared to going directly to Luke's house? (1 mark)

14 km instead of 10 km. ∴ 4 km further.

- States correct distance ✓

- (d) How much further is this, expressed as a percentage of the direct distance between the two houses?

(2 marks)

$$\begin{aligned} \frac{4}{10} \times 100 \\ = 40\% \text{ further} \end{aligned}$$

- Shows correct expression ✓

- Calculates percentage ✓

★ $\left(\frac{\text{Difference}}{\text{Original}} \times 100 \right) \%$

See next page

Question 7

(7 marks)

The weekly hours worked and hourly pay rates for three casual employees is shown in the table below.

Employee	A	B	C
Hours worked	10	12	20
Pay rate (\$)	18	10	15

- (a) Employee A always saves 15% of any amount they earn. How much should they save this week? (2 marks)

$$\text{Earnings} = 10 \times 18$$

$$= \$180$$

$$10\% \text{ of } 18$$

$$5\% \text{ of } 9$$

$$\therefore \text{They save } \$27.$$

- Calculates ✓
collect earnings
- Calculates correct amount of savings ✓

- (b) The employer would like to arrange the hours worked into matrix H and the pay rates into matrix R so that the matrix $H \times R$ will show the sum of the pay owing to all employees. Show how this can be done and calculate matrix $H \times R$. (2 marks)

$$H \begin{bmatrix} 10 & 12 & 20 \end{bmatrix} \times R \begin{bmatrix} 18 \\ 10 \\ 15 \end{bmatrix} = \begin{bmatrix} \$600 \end{bmatrix}$$

* Aiming for 1 value i.e. 1×1 matrix

$$\therefore H_{13} \times R_{31} = (HR)_{11}$$

- Shows H and R correctly ✓
- Calculates correct product. ✓

- (c) On their next birthday, the pay rate for employee C will increase by 3%. Calculate what their new hourly rate will be. (1 mark)

$$15 \times 0.03$$

$$0.15 \times 3$$

$$= 0.45$$

$$\therefore \text{New rate}$$

$$\text{is } \$15.45/\text{hour}$$

- Calculates ✓
correct rate

- (d) Calculate $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1.03 \end{bmatrix} \times R$ and explain what the resulting matrix shows. (2 marks)

$$\star \text{ Dimensions: } 3 \times 3 \times 3 = 3$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1.03 \end{bmatrix} \times \begin{bmatrix} 18 \\ 10 \\ 15 \end{bmatrix} = \begin{bmatrix} 18 \\ 10 \\ 15.45 \end{bmatrix}$$

- Calculates correct product ✓

- Appropriate explanation. ✓

The result shows pay rates after the pay rise for employee C.

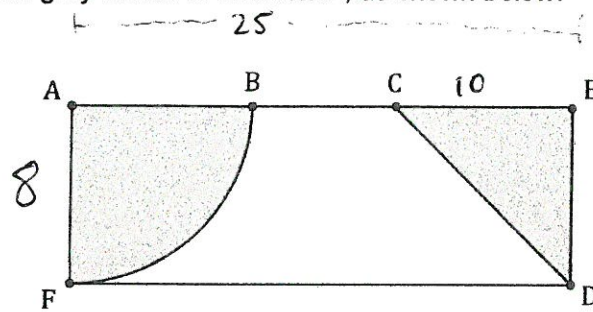
* Need to mention new pay rates or specify pay rates after employee C's birthday.

See next page

Question 8

(5 marks)

Rectangular logo $AEDF$, with $AF = 8$ cm and $AE = 25$ cm, has a white background with grey triangle CED and grey sector of circle ABF , as shown below.



Given that $2 \times AE = 5 \times CE$ and assuming that $\pi = 3$, determine what percentage of the area of the rectangular logo is grey.

$$50 = 5(CE)$$

$$\therefore CE = 10$$

$$\begin{aligned} \text{Area of Triangle} &= \frac{1}{2} \times 10 \times 8 \\ &= 40 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area of } \frac{1}{4} \text{ circle} &= \frac{\frac{1}{4} \times 8^2}{4} \\ &= \frac{3 \times 64}{4} \\ &= 48 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area of Rectangle} &= 8 \times 25 \\ &= 200 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \therefore \% \text{ Shaded} &= \frac{88}{200} \times 100 \\ &= 44\% \end{aligned}$$

- Calculates CE length ✓
- Calculates area of triangle ✓
- Calculates area of $\frac{1}{4}$ circle ✓
- Calculates area of rectangle ✓
- Calculates shaded area percentage. ✓

★ Use subheadings to clearly organise your working out

★ You should know the differences between formulae for:

- area of triangle
- area of sector / circle

→ Pythagoras' doesn't give you area
→ circumference is not area

End of questions

★ Read the Q.

$$\left(\frac{\text{Grey}}{\text{Total}} \times 100 \right) \%$$